

Mahmoud Adas

Software Engineer - Egypt

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About Me

C++ software engineer with 4 years building desktop dental CAD applications at Atomica.ai. Led the general-cad team and built internal tooling (SQLite code generation, library editor, launcher). Strong Qt/QML skills, with side interests in emulation, network protocols, and flight simulation.

Education

Cairo University, Faculty of Engineering | Egypt
B.S. in Computer Engineering

Sep 2015 - Jul 2021

Tools

- C++, C, Go, Python, Java, Rust
- Qt/QML, React, Spring, Android
- SQLite, Docker, Linux

Experience

C++ Software Engineer

Jun 2021 - Mar 2022, Jun 2023 - Now

Atomica.ai (360Imaging) - Giza, Egypt

- Led the general-cad-app team — prioritized feature work, coordinated cross-team efforts, and drove releases across multiple product cycles
- Maintained the frontend of dental implant planning and teeth segmentation applications, handling cross-team feature integration and bug fixes
- Built an ImGui/SQLite Library Editor enabling product managers to update dental catalog data (implants, abutments, materials) and release to AWS — eliminating app releases for data changes
- Created a SQLite-to-C++ code generator (ldb2cpp) producing type-safe database bindings, cutting boilerplate across multiple internal projects
- Contributed to the company's core C++ libraries (mn utility library, ndcore SDK) used across all product applications
- Developed an experimental Qt/QML launcher using libgit for file synchronization and a custom version-tracking file format — some components were later adopted by the main product suite

Software Engineering Intern · Cert

Jul - Sep 2020

Solectra - El Gouna, Egypt

- Maintained an **embedded system** that collects and analyses data from electric vehicles to report trips, total revenue, and energy
- Fixed embedded system bugs and improved EV trip/revenue prediction accuracy through data analysis and additional test coverage

Projects

360Launcher

Jan - Mar 2022

Qt/QML/C++ app to install/update/launch main products and integrate with them

Library Editor

Sep - Dec 2021

C++ (+ImGui) internal tool to modify/upload versioned metadata and information. Helps product managers update domain-specific logic without new app release.

DPSX-Sync

Jun - Aug 2021

- Qt/QML/C++ experimental Dropbox-like app to add sync and collaboration features
- Uses libgit and Gitea server for versioning and synchronization

C4IAN (Graduation Project) · Repo

Feb - Jul 2021

- Tactical Mobile ad-hoc network. Supports video/audio streaming and messaging
- ZHLS+ZID unicast, ODMRP multicast, and geo-based broadcast

- **Golang** router that integrates with the Linux network stack

OpenYSF · Repo

Nov 2021 - Present

A ysflight flight simulator clone, to explore game engine architectures. Backward compatible with ysflight assets.

Apex · Backend · Frontend · Android · Testing

Jan - Mar 2019

- Reddit clone, for web and android. Took role of **Devops** in team of 18
- Used Travis-CI for CI/CD, **GoogleCloud**, **Docker**, DockerCompose, and Datadog for monitoring

NesEmu · Repo

2019 - Present

NES emulator in C++ implementing a 6502 CPU with full instruction set, iNES ROM loading, PPU with native NES palette and pattern table rendering, and an ImGui debugger featuring memory inspection, live disassembly, and tile/palette viewers. Built with SFML.

Koos · Repo · Rust Rewrite

Jun 2019

- Decentralized social media app, built on Fluence. AssemblyScript/Rust for server, React for frontend
- Won a Bitcoin bounty for the best app that uses Fluence's technology

Match Reservation System · Repo

Jan - Feb 2021

Web application for Egyptian Premier League for seats reservation. Used **MongoDB**, **Nodejs** and React

Mini-Search-Engine · Repo

Feb - Mar 2020

Text and image search with trends detection. Used Java Spring and **React**

Cartoonization · Repo

Oct - Dec 2019

- Snapchat Clone written in Python draws stickers on live recordings of faces
- Implemented GUI, rendering stickers, and HOG algorithm
- Used Viola-jones/Hog to detect faces, and multiple **image processing** techniques to filter the input

RetroSoccer · Repo

Jan - Mar 2019

Multiplayer soccer game entirely in x86 IA32 MASM assembly with chat (serial communication)

Simple PDP--11 · Repo

Oct - Dec 2019

- Design and simulation of a CPU architecture similar to *PDP--11* in VHDL and Vunit for test benches
- Designed and implemented automated tests for the CPU

TheMistake · Repo

Jan - Mar 2019

- Maze solver car using flood fill algorithm on TIVA-C μ C/Arduino-Uno with ultrasonic and IR sensors
- Implemented maze-solving and tested it